DIGITAL TWIN ENGINEERING · DT-G04

CereFlow-*DT*

A digital twin that watches **the brain's blood supply**, live.



69%

When one of our volunteers stood up, blood flow in the right side of the brain fell to **69% of normal** within seconds.

Our twin noticed. Would your monitor?



Monitors show waveforms. Clinicians need *answers*.

1

Is enough blood reaching each side of the brain?

Compared with this person's own baseline, not a textbook average.

2

Is the brain still holding its flow steady?

Cerebral autoregulation: does flow just follow blood pressure?

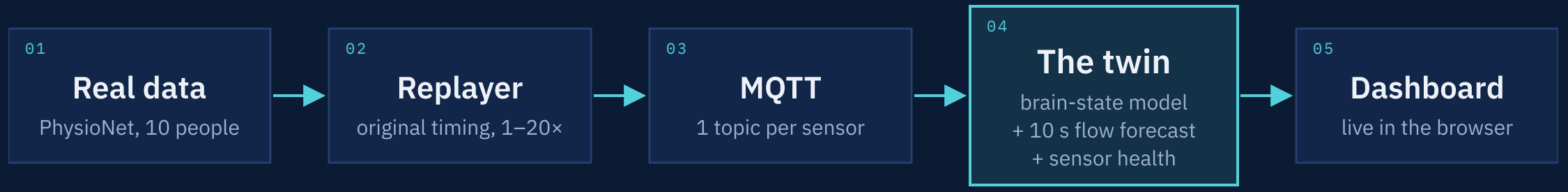
3

Is flow about to drop, and can we trust the sensors?

Early warning, broken-sensor detection, and a model that says when it's unsure.

FOR | Clinicians running stand-up / tilt tests | Neurocritical care after an aneurysm bleed | Engineers tuning the monitoring

Real recordings in, *live brain state* out



Replayer and twin talk only through the broker: swap the replay for a live patient feed and nothing else changes.

52
real sensor streams of 4 types: Doppler, blood pressure, breathing, NIRS

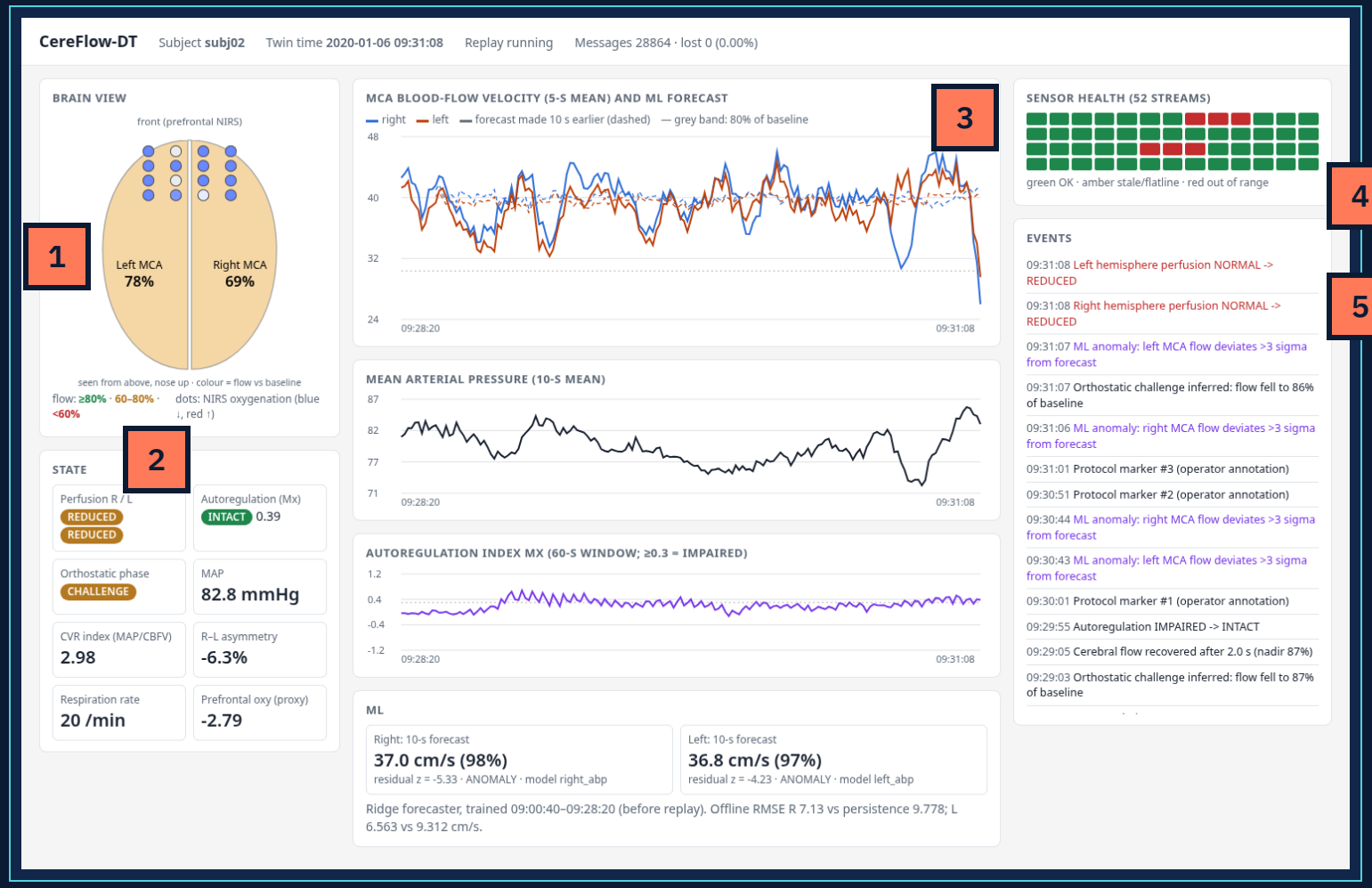
64k
time steps across 10 healthy adults, ~33 min each

4
derived states: perfusion, autoregulation, posture phase, sensor health

10 s
flow forecast per hemisphere, trained only on data before the replay

DATA · PhysioNet "Cardiorespiratory response to orthostatic stress" v1.0 · Mukli et al., Front. Physiol. 2021

The moment subject 02 *stands up*



1. **Brain map** left side at 78%, right at 69% of this person's baseline.
2. **State** both sides REDUCED, phase CHALLENGE, autoregulation still INTACT.
3. **Forecast** the drop breaks the 10-second forecast, so it's flagged as an anomaly.
4. **Sensor health** red cells are NIRS channels with bad contact, flagged automatically.
5. **Event log** every state change, time-stamped.

Measured, *not promised*

19/20

hemispheres where the 10 s forecast beats "no change"

median 27.5% lower error

0

messages lost at up to 20× speed

replay timing within 0.4% of target

100%

of known faulty sensors flagged

auto-switches model when blood pressure fails

6/10

stand-ups detected, 3.7–16 s after standing

with hand-set thresholds; tuning is next

Every number re-checked by one script `python verify_report.py` **29 / 29 OK**

05 · WHAT'S NEXT

From toy twin to *bedside twin*

01 Tune alarm thresholds on held-out subjects to lift 6/10 detection

02 Add a live "coupling network" view of heart, breathing and brain, after Mukli et al. 2021

03 Validate on patient data after aneurysm bleeds

ONE ASK

The dataset has elapsed seconds, not dates. We anchor it to a fixed, configurable start time.

Can we confirm this counts as an absolute timestamp?

CereFlow-DT: *know the brain is getting enough blood*, before it isn't.